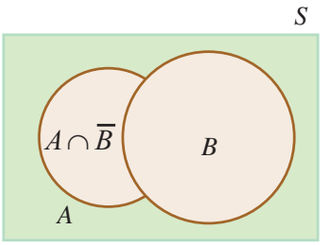
Theorem 12.6 (Addition theorem on probability)

If A and B are any two events, then

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$



Proof

From the diagram,

$$A \cup B = (A \cap \overline{B}) \cup B$$

$$P(A \cup B) = P\left[(A \cap \overline{B}) \cup B\right]$$

$$= P(A \cap \overline{B}) + P(B) \quad \text{(since } (A \cap \overline{B}) \text{ and } B \text{ are mutually exclusive)}$$

$$= [P(A) - P(A \cap B)] + P(B)$$

Therefore, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ ■

Note 12.4

The above theorem can be extended to any 3 events.

(i) $P(A \cup B \cup C) = \{P(A) + P(B) + P(C)\}$

$$- \{P(A \cap B) + P(B \cap C) + P(C \cap A)\} + P(A \cap B \cap C)\}$$

(ii) $P(A \cup B \cup C) = 1 - P(\overline{A \cup B \cup C}) = 1 - P(\overline{A} \cap \overline{B} \cap \overline{C})$

Example 12.14

Given that $P(A) = 0.52$, $P(B) = 0.43$, and $P(A \cap B) = 0.24$, find

(i) $P(A \cap \overline{B})$ (ii) $P(A \cup B)$ (iii) $P(\overline{A} \cap \overline{B})$ (iv) $P(\overline{A} \cup \overline{B})$.

Solution

(i) $P(A \cap \overline{B}) = P(A) - P(A \cap B)$

$$= 0.52 - 0.24 = 0.28$$

$$P(A \cap \overline{B}) = 0.28.$$

(ii) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$= 0.52 + 0.43 - 0.24$$

$$P(A \cup B) = 0.71.$$

(iii) $P(\overline{A} \cap \overline{B}) = P(\overline{A \cup B})$ (By de Morgan's law)

$$= 1 - P(A \cup B)$$

$$= 1 - 0.71 = 0.29.$$

(iv) $P(\overline{A} \cup \overline{B}) = P(\overline{A \cap B})$ (By de Morgan's law)

$$= 1 - P(A \cap B) = 1 - 0.24$$

$$= 0.76.$$

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